



MERN STACK + AI ENGINEERING BOOTCAMP

Week 2 — Assignments 2

Advanced Interface Engineering & Structural Layout Workspace

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Course	MERN Stack + AI Engineering Bootcamp
Module	Week 2 — Box Mapping & Wireframe Documentation
Date	5 june 2026
Status	Completed

ASSIGNMENT 02

Box Mapping & Wireframe Documentation

CSS Theoretical Knowledge Base & Repository Architecture Layout

Reference Page	Shopamp! — Clothing Store Landing Page
Source Colors	--clr-primary: #3d9cd2 --clr-dark: #1e293b --clr-accent: #00b5e2
Total Sections	13 Sections Analysed
Status	Completed

CONTENTS:

HTML DESIGN SYSTEM — Sementic tags & layout structure

NO.	SECTION	HTML CLASS / TAG
01	NAVBAR	<header class='navbar'>
02	HERO SECTION	<section class='hero'>
03	ABOUT US	<section class='about'>
04	SALE BANNER	<section class='sale-promo-banner'>
05	FEATURES / STORE USP _s	<section class='features'>
06	OUR PRODUCTS	<section class='products'>
07	NEWSLETTER / SUBSCRIBE	<section class='modern-newsletter-section'>
08	SHOP BANNERS	<section class='shop-banners'>
09	LATEST NEWS	<section class='news'>

10	LAST COLLECTION / GALLERY	<section class='instagram'>
11	TESTIMONIALS	<section class='testimonials'>
12	OUR LOCATION	<section class='location'>
13	FOOTER	<footer class='footer'>

CONTENTS:

CSS DESIGN SYSTEM — Global Variables & Breakpoints

Custom Properties (:root)

CSS VARIABLE	VALUE	USAGE
--clr-primary	#3d9cd2	Main brand color — nav logo, buttons, headings, icons, links
--clr-primary-hover	#2a7fb5	Darker primary on hover states
--clr-accent	#00b5e2	Accent blue (defined, minimal direct use)
--clr-dark	#1e293b	Main dark text color — headings, nav links
--clr-text-gray	#555555	Body text — card descriptions, footer desc
--clr-text-light	#888888	Muted text — card labels, footer copyright
--clr-bg-light	#f0f2f7	Section backgrounds — about, features, news, footer
--clr-bg-app	#f4f6f9	Newsletter section bg
--clr-white	ffffff	White backgrounds
--clr-border	#cbd5e1	Input borders, dividers
--clr-muted	#64748b	Muted medium-gray text
--nav-height	64px	Fixed navbar height — used for hero margin-top
--radius-sm	4px	Small border radius — cards, buttons, icons

--radius-lg	30px	Pill radius — Buy Now, Watch Now buttons
--font-display	Montserrat	Headings, nav, logo, buttons
--font-body	Open Sans	Body text, descriptions, paragraphs
--shadow-sm	0 2px 12px rgba(0,0,0,0.12)	Navbar scrolled shadow
--shadow-hover	0 20px 38px rgba(0,0,0,0.22)	News card hover shadow
--transition-fast	0.2s cubic- bezier(0.4,0,0.2,1)	Quick interactions
--transition-premium	0.4s cubic- bezier(0.25,1,0.5,1)	Premium elastic easing
--container-width	1100px	Max width of .container
--layout-max-width	1140px	Max width for contact/location sections
--grid-gap-main	24px	Instagram gallery grid gap

Responsive Breakpoints (@media)

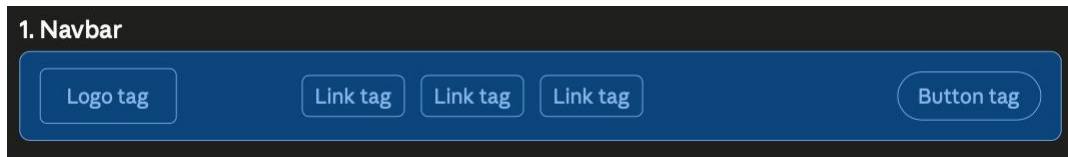
BREAKPOINT	CONDITION	KEY CHANGES
Tablet	width <= 900px	.three-col → 1-col; features-layout → 1-col; footer-grid → 2-col; insta-grid → 2-col; banner → 1-col
Tablet Contact	width <= 768px	contact-split-header → 1-col; location-grid → 1-col
Mobile	width <= 640px	hamburger shown; nav-links → fixed full-width slide-down; two-col → 1-col; newsletter-form → column; footer → 1-col

1	NAVBAR <code><header class='navbar' id='navbar'></code>	LAYOUT
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► ① WIREFRAME DIAGRAM

NAVBAR — Full Width Sticky height: 64px position: fixed	
c shopamp! (LOGO) font: Montserrat 700 color: #3d9cd2	HOME LIVE DEMO LIVE DEMO BLOCKS [Buy Now]

Reference Image:



► ② BOX TREE — Parent → Child Hierarchy

[header.navbar] NAVBAR CONTAINER
├── [div.nav-container] → max-width: 1200px flex layout
├── [a.logo] → "shopamp!" color: #3d9cd2 font: Montserrat
├── [nav.nav-links#main-menu] → display: flex gap: 28px
├── [a.nav-link.active] → "HOME" font-size: 0.82rem font-weight: 600
├── [div.nav-dropdown] → "LIVE DEMO" + dropdown-menu
├── [div.nav-dropdown] → "LIVE DEMO BLOCKS" + dropdown-menu
└── [a.btn-buy] → "Buy Now" bg: #3d9cd2 border-radius: 30px
└── [button.hamburger] → mobile toggle display: none (desktop)

► ③ HTML TAGS — Element Reference Table

ELEMENT	HTML TAG	CLASS / ID	SOURCE NOTES
NAVBAR CONTAINER	<header>	.navbar #navbar	position: fixed; top: 0; z-index: 1000; height: 64px
NAV WRAPPER	<div>	.nav-container	max-width: 1200px; display: flex; align-items: center; justify-content: space-between
LOGO	<a>	.logo	color: #3d9cd2; font: Montserrat 700; font-size: 1.5rem; letter-spacing: -0.5px
NAV LINKS	<nav>	.nav-links #main-menu	display: flex; align-items: center; gap: 28px
NAV LINK	<a>	.nav-link .active	font: Montserrat 600; 0.82rem; letter-spacing: 0.5px; color: #1e293b
DROPDOWN WRAPPER	<div>	.nav-dropdown	position: relative; hover reveals .dropdown-menu
DROPDOWN TOGGLE	<button>	.dropdown-toggle	aria-haspopup: true; aria-expanded; background: transparent; border: none
DROPDOWN MENU	<div>	.dropdown-menu	position: absolute; min-width: 160px; border-top: 2px solid #3d9cd2; opacity: 0 → 1 on hover
CTA BUTTON	<a>	.btn-buy	bg: #3d9cd2; color: #fff; border-radius: 30px; padding: 9px 22px; font: Montserrat 600
HAMBURGER	<button>	.hamburger #hamburger	display: none (desktop) → flex (mobile ≤640px); aria-expanded toggle

► ④ CSS PROPERTIES — with Code Snippets

PROPERTY	VALUE	PURPOSE	CODE SNIPPET
position	fixed	Stays on top on scroll	position: fixed; inset: 0 0 auto 0;

height	64px (<code>--nav-height</code>)	Fixed compact navbar height	<code>height: var(--nav-height);</code>
background	<code>rgba(255,255,255,0.97)</code>	Near-opaque white with blur	<code>backdrop-filter: blur(8px);</code>
z-index	1000	Above all page content	<code>z-index: 1000;</code>
box-shadow (scrolled)	0 2px 12px <code>rgba(0,0,0,0.12)</code>	Shadow on scroll via <code>.scrolled</code> class	<code>.navbar.scrolled { box-shadow: var(--shadow-sm); }</code>
gap (nav-links)	28px	Space between nav items	<code>.nav-links { display: flex; gap: 28px; }</code>
font (logo)	Montserrat 700 1.5rem	Brand logo typography	<code>.logo { font-family: var(--font-display); font-weight: 700; }</code>
color (logo)	<code>#3d9cd2</code> (<code>--clr-primary</code>)	Brand primary color	<code>.logo { color: var(--clr-primary); }</code>
background (btn-buy)	<code>#3d9cd2</code> → <code>#2a7fb5</code> hover	Primary CTA with hover state	<code>.btn-buy:hover { background: var(--clr-primary-hover); transform: translateY(-1px); }</code>
border-radius (btn)	30px (<code>--radius-lg</code>)	Pill-shaped CTA button	<code>.btn-buy { border-radius: var(--radius-lg); }</code>
dropdown border-top	2px solid <code>#3d9cd2</code>	Brand accent top border on dropdown	<code>.dropdown-menu { border-block-start: 2px solid var(--clr-primary); }</code>

► ⑤ DESIGN STRATEGY

STRATEGY	WHY USED	SOURCE EVIDENCE
Fixed + Backdrop Blur	Keeps nav accessible as user scrolls long page	<code>position: fixed; backdrop-filter: blur(8px);</code> JS adds <code>.scrolled</code> class on <code>scrollY > 60</code>
Flexbox Layout	Spaces logo, nav links, CTA cleanly in one row	<code>.nav-container { display: flex; justify-content: space-between; align-items: center; }</code>
Single Pill CTA	One action (Buy Now) focuses	<code>.btn-buy { border-radius: var(--radius-</code>

	conversion intent	lg); padding: 9px 22px; }
ARIA Dropdowns	Accessible keyboard/screen-reader navigation	aria-haspopup, aria-expanded, aria-controls — JS toggles these on click
Mobile Hamburger	Collapses nav at ≤ 640px breakpoint	@media (width <= 640px) { .hamburger { display: flex; } .nav-links { position: fixed; transform: translateY(-110%); } }

2

HERO SECTION `<section class='hero' id='home'>`

LAYOUT

► ① WIREFRAME DIAGRAM

HERO — Full Viewport | position: relative | height: 100vh | margin-top: 64px

BACKGROUND IMAGE (img.hero-bg) object-fit: cover object-position: center top

OVERLAY rgba(240,240,235,0.45)

h1 "Clothing Store" font: Montserrat 700 3.8rem color: #1e293b

p "Clothing Website Template" color: #3d9cd2

Reference Image:



2. Hero Section

Background Image tag

h1 Heading tag

Paragraph tag

Button tag

Button tag

► ② BOX TREE — Parent → Child Hierarchy

[section.hero#home] HERO CONTAINER	position: relative height: 100vh
—— [div.hero-overlay]	→ position: absolute; inset: 0; background: rgba(240,240,235,0.45)
—— [img.hero-bg]	→ src="Image.jpg" position: absolute object-fit: cover object-position: center top
—— [div.hero-content]	→ position: absolute inset-block-end: 80px inset-inline-start: 48px
—— [h1.hero-title]	→ "Clothing\nStore" font: Montserrat 700 3.8rem
—— [p.hero-subtitle]	→ "Clothing Website Template" color: #3d9cd2 font-weight: 600

► ③ HTML TAGS — Element Reference Table

ELEMENT	HTML TAG	CLASS / ID	SOURCE NOTES
HERO CONTAINER	<section>	.hero #home	position: relative; height: 100vh; min-height: 560px; overflow: hidden
BACKGROUND IMAGE		.hero-bg	src="Image.jpg"; position: absolute; inset: 0; width/height: 100%; object-fit: cover; object-position: center top
OVERLAY	<div>	.hero-overlay	position: absolute; inset: 0; background: rgba(240,240,235,0.45) — light warm tint
HERO CONTENT	<div>	.hero-content	position: absolute; inset-block-end: 80px; inset-inline-start: 48px
HEADLINE	<h1>	.hero-title	"Clothing\nStore" — font: Montserrat 700; 3.8rem; line-height: 1.15; color: #1e293b
SUBTITLE	<p>	.hero-subtitle	color: #3d9cd2; font-weight: 600; letter-spacing: 0.5px; margin-top: 10px

4.CSS PROPERTIES — with Code Snippets

PROPERTY	VALUE	PURPOSE	CODE SNIPPET
height	100vh min-height: 560px	Full viewport hero	height: 100vh; min-height: 560px;
margin-top	64px (---nav-height)	Clears fixed navbar	margin-block-start: var(---nav-height);
object-fit (img)	cover	BG image fills container	img.hero-bg { object-fit: cover; position: absolute; inset: 0; }
background (overlay)	rgba(240,240,235,0.45)	Light warm tint (not dark!)	.hero-overlay { background: rgba(240,240,235,0.45); }
font-size (h1)	3.8rem Montserrat 700	Large impactful headline	.hero-title { font-size: 3.8rem; font-weight: 700; line-height: 1.15; }
color (h1)	#1e293b (---clr-dark)	Dark text on light overlay	.hero-title { color: var(---clr-dark); }
color (subtitle)	#3d9cd2 (---clr-primary)	Brand primary accent	.hero-subtitle { color: var(---clr-primary); font-weight: 600; }
position (content)	absolute bottom: 80px left: 48px	Bottom-left content placement	.hero-content { position: absolute; inset-block-end: 80px; inset-inline-start: 48px; }

5.DESIGN STRATEGY

STRATEGY	WHY USED	SOURCE EVIDENCE
Full-Viewport Hero	Instantly communicates fashion brand identity	height: 100vh; min-height: 560px — takes entire screen real estate
Light Warm Overlay (not dark)	Keeps brand feel clean and airy — fashion aesthetic	rgba(240,240,235,0.45) — very light, not the typical dark overlay
Bottom-Left Content	Asymmetric — feels editorial/magazine	inset-block-end: 80px; inset-inline-start: 48px — NOT centered
loading: eager	Hero image is LCP — load immediately	img.hero-bg loading="eager" — prioritises above-fold performance

3

ABOUT US <section class='about section-pad' id='about'>

LAYOUT

► ① WIREFRAME DIAGRAM

ABOUT US — 3-Column Card Grid | padding-block: 80px | bg: white

h2 "About Us" color: #3d9cd2 font: Montserrat 700 2.2rem text-align: center margin-bottom: 48px

💡💡 Prices — "Best prices" card

💡💡 Trends — "Newest trends" card

💡💡 Collections — "Previous collections" card

Reference Image:

About Us



PRICES

Best prices

In our store, you would find the best prices in the city. Every month we provide our customers with discounts and sales on certain categories of goods.

[Learn more >](#)



TRENDS

Newest trends

We always follow trends and try to collect the latest and most fashionable clothes in our shop. We have magazines, in which you can check out styles and colors.

[Learn more >](#)



COLLECTIONS

Previous collections

It's possible to buy the best pieces of clothes from previous collections with huge discounts. Every Saturday we provide our customers with sales.

[Learn more >](#)

3. About Us Section



► ② BOX TREE — Parent → Child Hierarchy

[section.about.section-pad#about]	ABOUT CONTAINER	padding-block: 80px
— [div.container]	→	max-width: 1100px margin: auto padding-inline: 24px
— [h2.section-title]	→	"About Us" color: #3d9cd2 text-align: center
— [div.cards-grid.three-col]	→	display: grid grid-template-columns: repeat(3,1fr) gap: 32px
— [article.card]	→	CARD 1 — Prices
— [div.card-img-wrap]	→	height: 200px overflow: hidden
— [img]	→	Unsplash URL object-fit: cover transition: scale(1.05) on hover
— [p.card-label]	→	"Prices" font: 0.78rem uppercase letter-spacing: 1px color: #888
— [h3.card-heading]	→	"Best prices" color: #3d9cd2
— [p.card-text]	→	description color: #555
— [a.card-link]	→	"Learn more >" aria-label
— [article.card]	→	CARD 2 — Trends
— [article.card]	→	CARD 3 — Collections

► ③ HTML TAGS — Element Reference Table

ELEMENT	HTML TAG	CLASS / ID	SOURCE NOTES
ABOUT CONTAINER	<section>	.about .section-pad #about	padding-block: 80px; background: white
INNER WRAPPER	<div>	.container	max-width: 1100px (--container-width); margin-inline: auto; padding-inline: 24px
SECTION TITLE	<h2>	.section-title	color: #3d9cd2; text-align: center; font: Montserrat 700; 2.2rem; margin-bottom: 48px
CARD GRID	<div>	.cards-grid .three-col	display: grid; grid-template-columns: repeat(3,1fr); gap: 32px

CARD	<article>	.card	Semantic article element; opacity/translateY animation via IntersectionObserver
IMAGE WRAP	<div>	.card-img-wrap	height: 200px; overflow: hidden; border-radius: 4px
CARD IMAGE		—	object-fit: cover; height: 100%; transition: scale(1.05) on .card:hover
CARD LABEL	<p>	.card-label	"Prices/Trends/Collections" — 0.78rem; text-transform: uppercase; color: #888888
CARD HEADING	<h3>	.card-heading	color: #3d9cd2; font: Montserrat 600; 1.1rem; margin-block: 6px 10px
CARD TEXT	<p>	.card-text	font-size: 0.88rem; color: #555555; line-height: 1.65
CARD LINK	<a>	.card-link	"Learn more >" — color: #555 → #3d9cd2 on hover; aria-label for accessibility

► ④ CSS PROPERTIES — with Code Snippets

PROPERTY	VALUE	PURPOSE	CODE SNIPPET
display	grid	3-column card layout	.cards-grid.three-col { grid-template-columns: repeat(3,1fr); gap: 32px; }
padding-block	80px (section-pad)	Consistent section spacing system	.section-pad { padding-block: 80px; }
card img hover	scale(1.05)	Hover zoom effect on card image	.card:hover .card-img-wrap img { transform: scale(1.05); }
card animation	opacity 0 → 1 translateY 24px→0	IntersectionObserver scroll animation	.card { opacity: 0; transform: translateY(24px); } .card.visible { opacity: 1; transform: translateY(0); }